

Nathaniel Felsky-Deelstra

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Summary

First-year Computer Engineering student with a passion for embedded systems, electronics, and team-based engineering design. Strong foundation in C/C++ and hardware prototyping, with practical experience in sensor integration, interference mitigation, and leadership. Seeking a summer engineering position to apply problem-solving, initiative, and technical skills.

Education

Queen's University, BAsc Electrical and Computer Engineering

Clubs and Teams

Simulation Lead, aQuatonomous May 2026 – Present

- Created simulations of the teams ASV on Gazebo using ROS 2 Jazzy
- Led a group of students to improve existing systems and testing capabilities allowing for other software teams to verify their results

Electrical Team Member, aQuatonomous September 2025 – May 2026

- Developed interference-mitigation strategies for the telemetry subsystem using magnetic field calculations.
- Designed sensor, radio, and wiring layout to reduce noise and improve signal reliability.
- Conducted research on Faraday cages and magnetic field attenuation to inform design decisions.

Elected Year Representative, Queen's ECE Executive Club September 2025 – Current

- Acted as liaison for the 1st year CE and the ECE exec team, gathering feedback and communicating issues.
- Designed a first-year program patch and supported branding.
- Assisted in planning and running community-building events, including ECE Night and intramural activities.

Experience

Social & Economic Policy Researcher, Canadian Labour Congress June 2023 – July 2026,

- Researched modular construction tech and AI focusing on the impacts to society and future growth of the tech.

Projects

Personal Website

- Learned web dev skills and worked with data analytics to improve site SEO.

Modern Slavery Hot Line

- Created a hot line to educate at risk workers about forced labour and their rights.
- Pipeline consisting of Twilio, Claude, Google Speech, Resend.
- Optimized responses to under 3 seconds, while maintaining high quality and good user experience.

Water Pre-Treatment Plant

- Designed a turbidity-reduction system, achieving more than 2500 NTU's improvement using coagulation and press filtration in under 8 minutes.
- Designed mechanical parts using CAD and calculated gear ratios for accurate meshing and motor speed.
- Implemented Arduino-based automation and wiring for pump and sensor control.

Skills

Languages: C, C++, C#, Python

Software: Autodesk Fusion, Jupyter Notebook, Excel, Word, Visual Studio Code, Arduino IDE, Gazebo, ROS 2